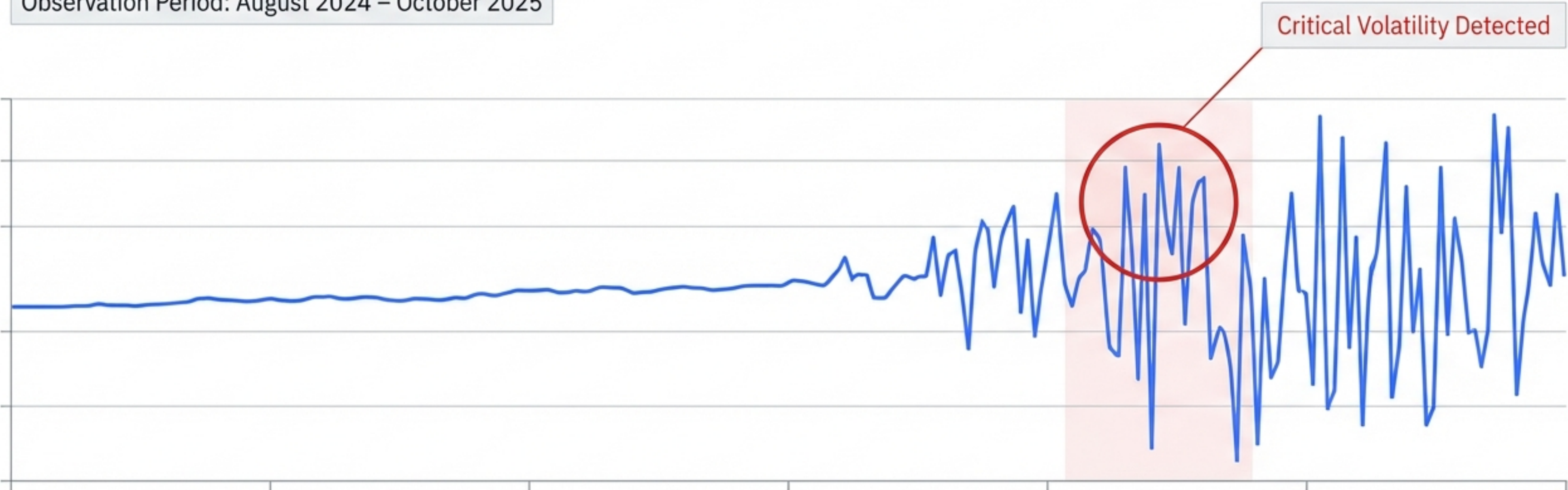


NEO4J SYSTEM HEALTH & ML PERFORMANCE REVIEW

Diagnostic analysis of resource utilisation, model fidelity, and stability.

Observation Period: August 2024 – October 2025



ANALYSIS SCOPE: 14 MONTHS

DATA VOLUME: 275+ METRIC EVENTS

DOMAINS: INFRASTRUCTURE & ML OUTPUT

EXECUTIVE DIAGNOSIS: HIGH VOLATILITY WITH CRITICAL ANOMALIES

THROUGHPUT



Peak: ~332 ops

System demonstrates consistent peak performance capability.
Not resource constrained.

INFRASTRUCTURE



CPU Spikes: >1000%

Severe bursting events detected.
Multi-threaded bottlenecks.
GPU logging corruption likely.

MODEL HEALTH



Accuracy Variance: 0% - 89%

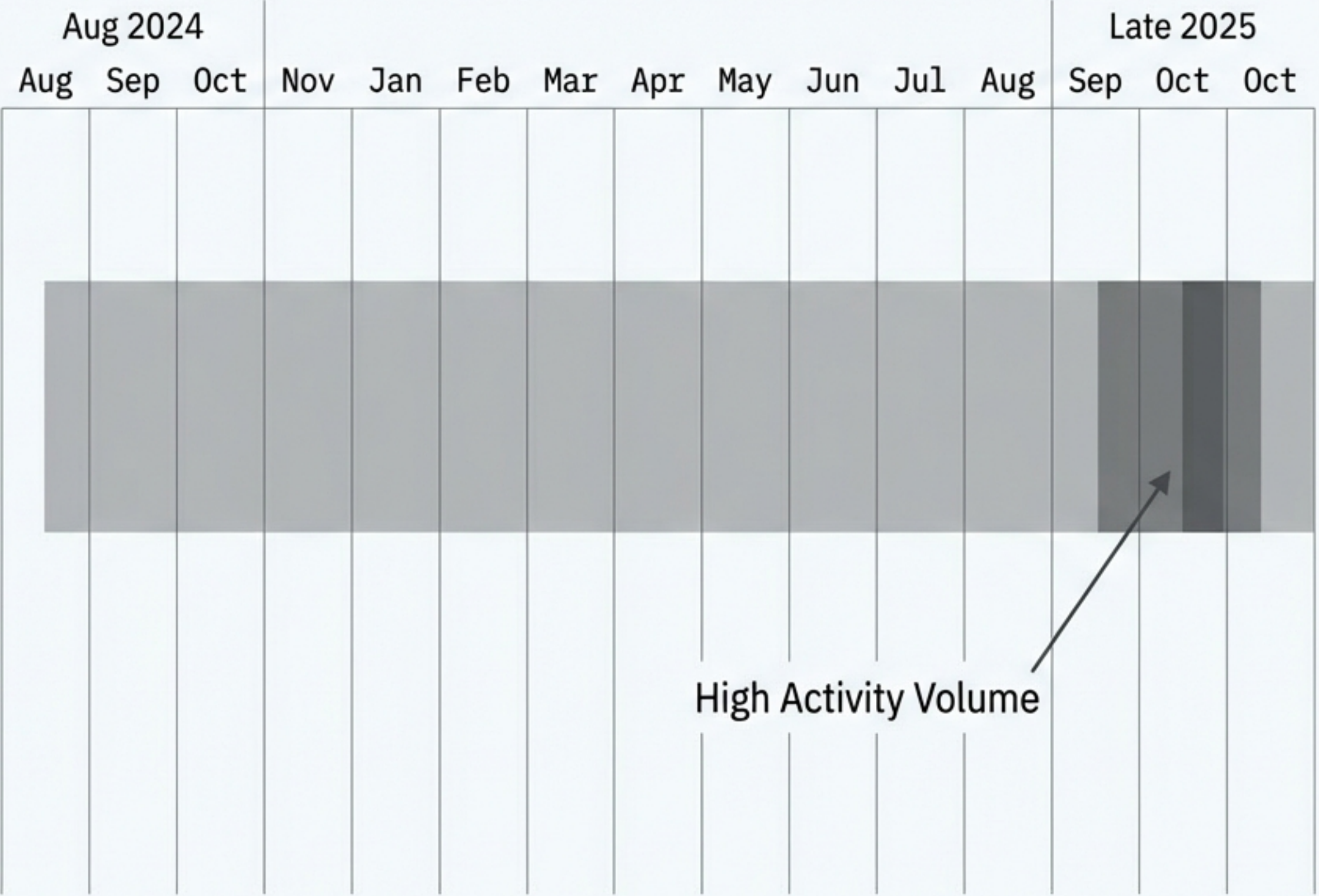
Highly unstable. Significant
'wasted compute' on models
returning 0% accuracy.

KEY INSIGHT: The system is not resource-constrained but rather unoptimised. We are observing 'burst' events where CPU usage exceeds 1000%, correlated with specific batch processing windows.

CLINICAL SCOPE & METHODOLOGY

Analysis based on raw extraction from 'neo4j_metrics.csv'. The dataset covers a 14-month operational window.

- [x] INFRASTRUCTURE VITAL SIGNS:
cpu_usage, memory_usage, gpu_utilization
- [x] SYSTEM REFLEXES:
throughput, latency
- [x] COGNITIVE OUTPUT:
accuracy, loss, error_rate

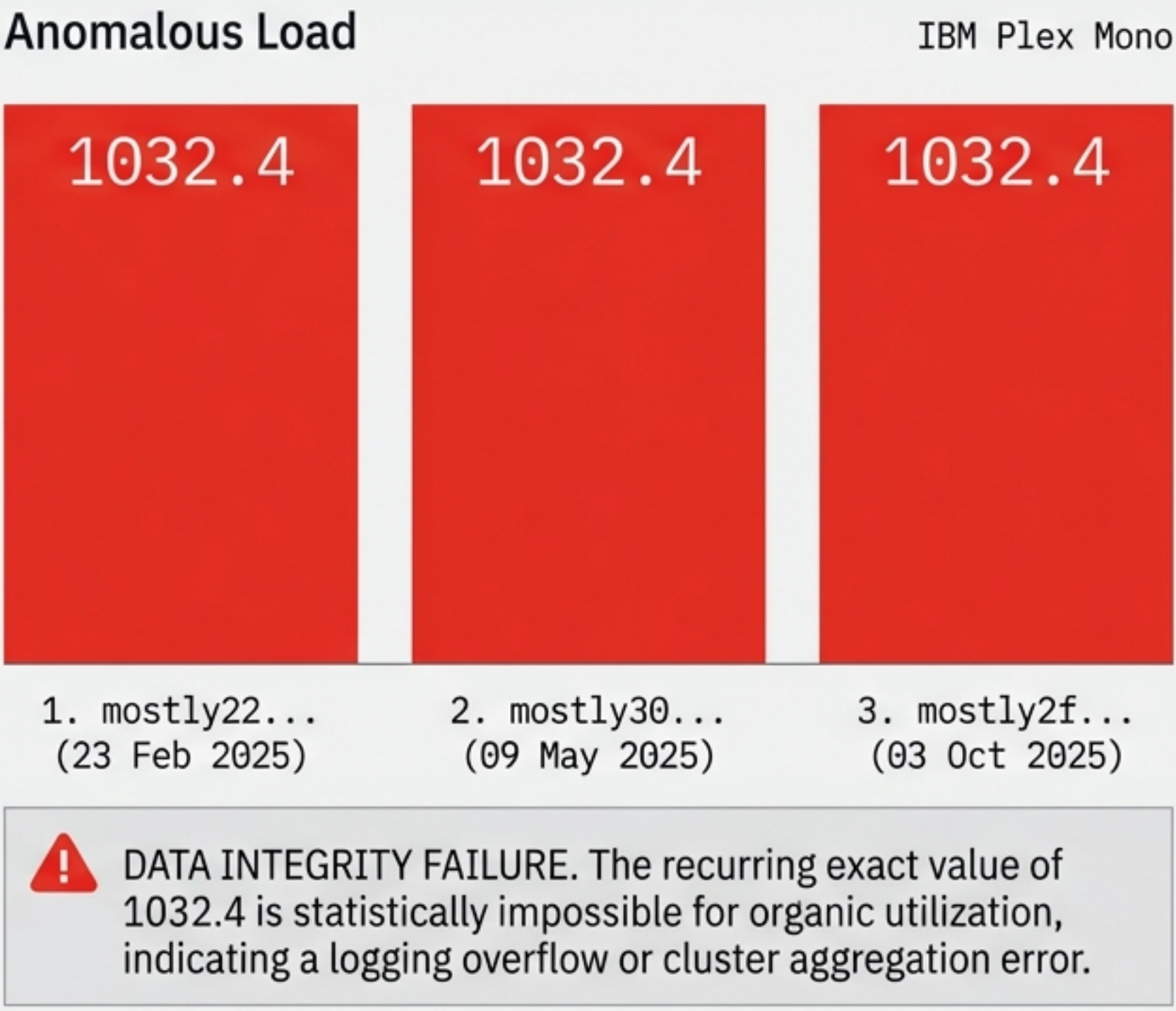
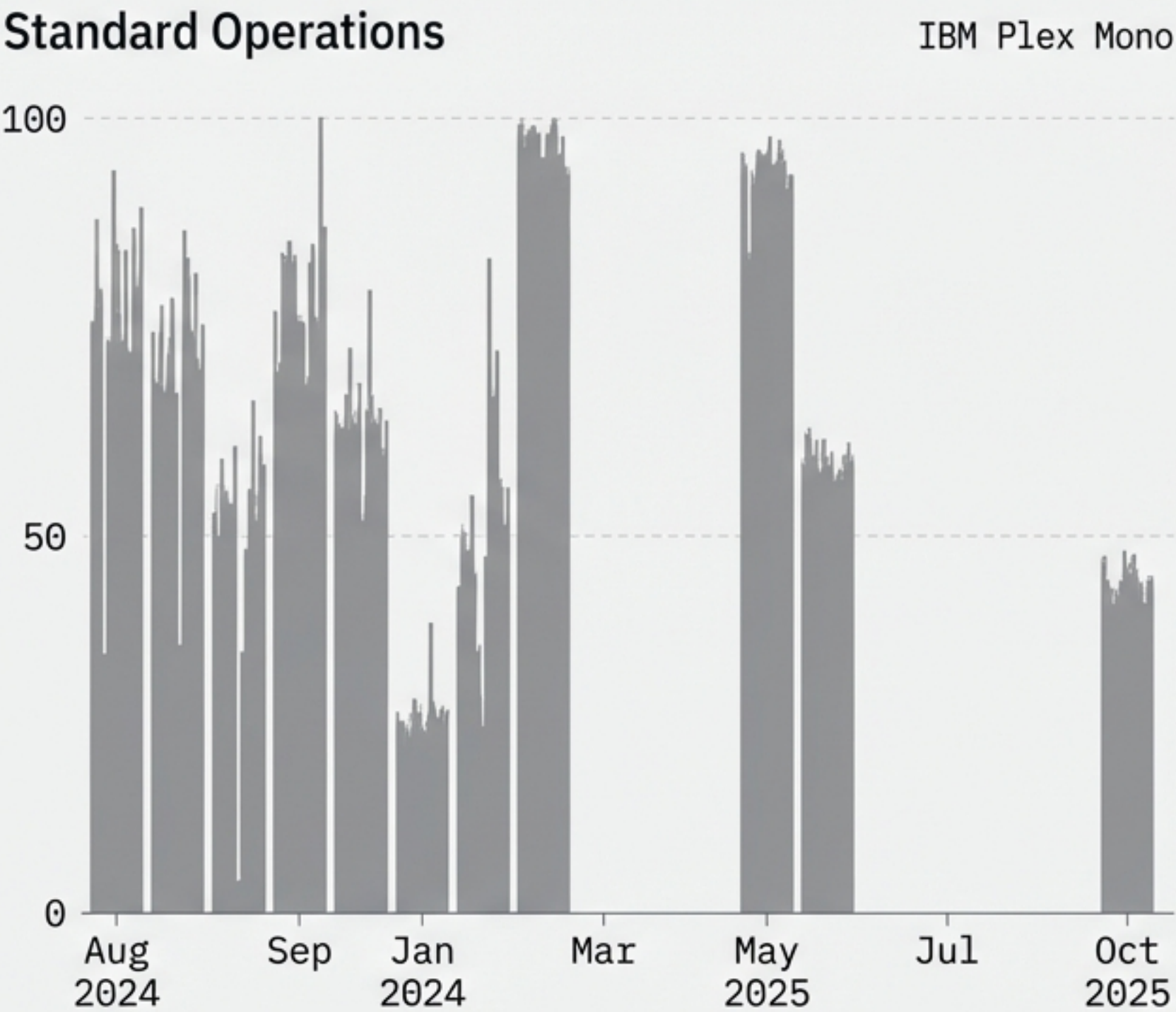


INFRASTRUCTURE VITALS: CPU TACHYCARDIA

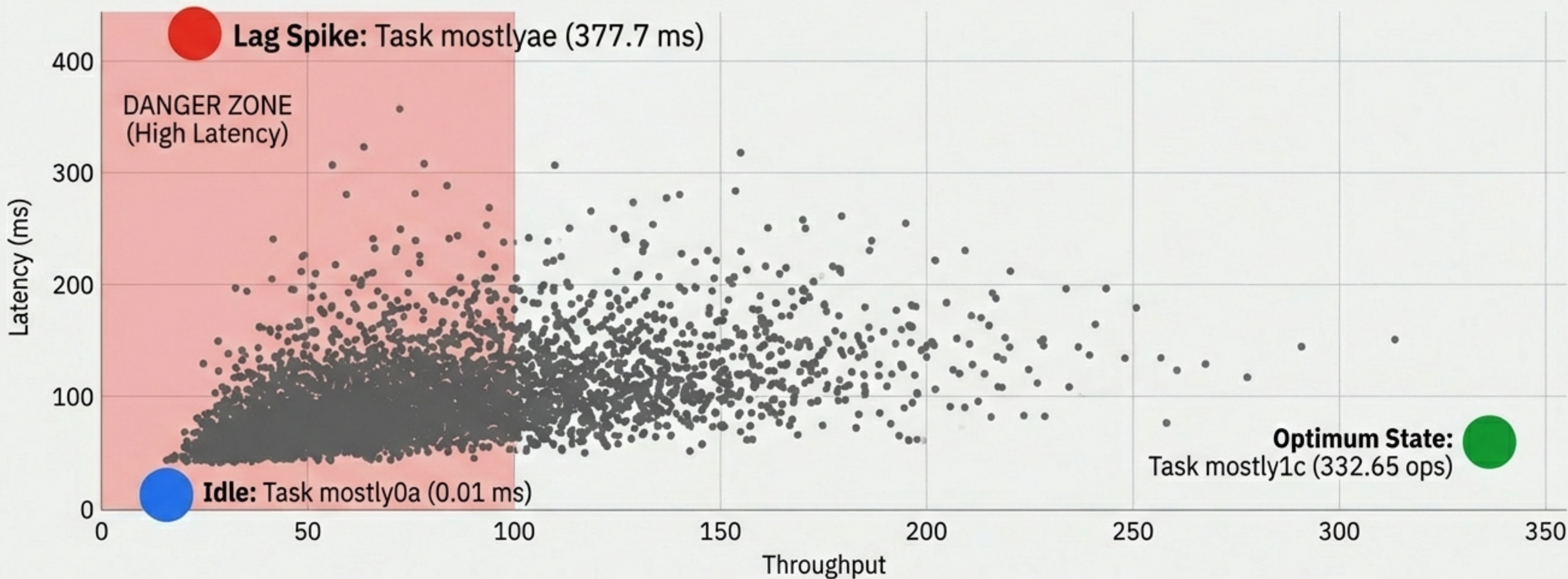


INSIGHT: Values exceeding 100% indicate aggressive multi-threaded parallelism, likely GDS projection algorithms seizing all available cores. This behaviour creates resource starvation risks.

INFRASTRUCTURE VITALS: THE GPU '1032' ANOMALY

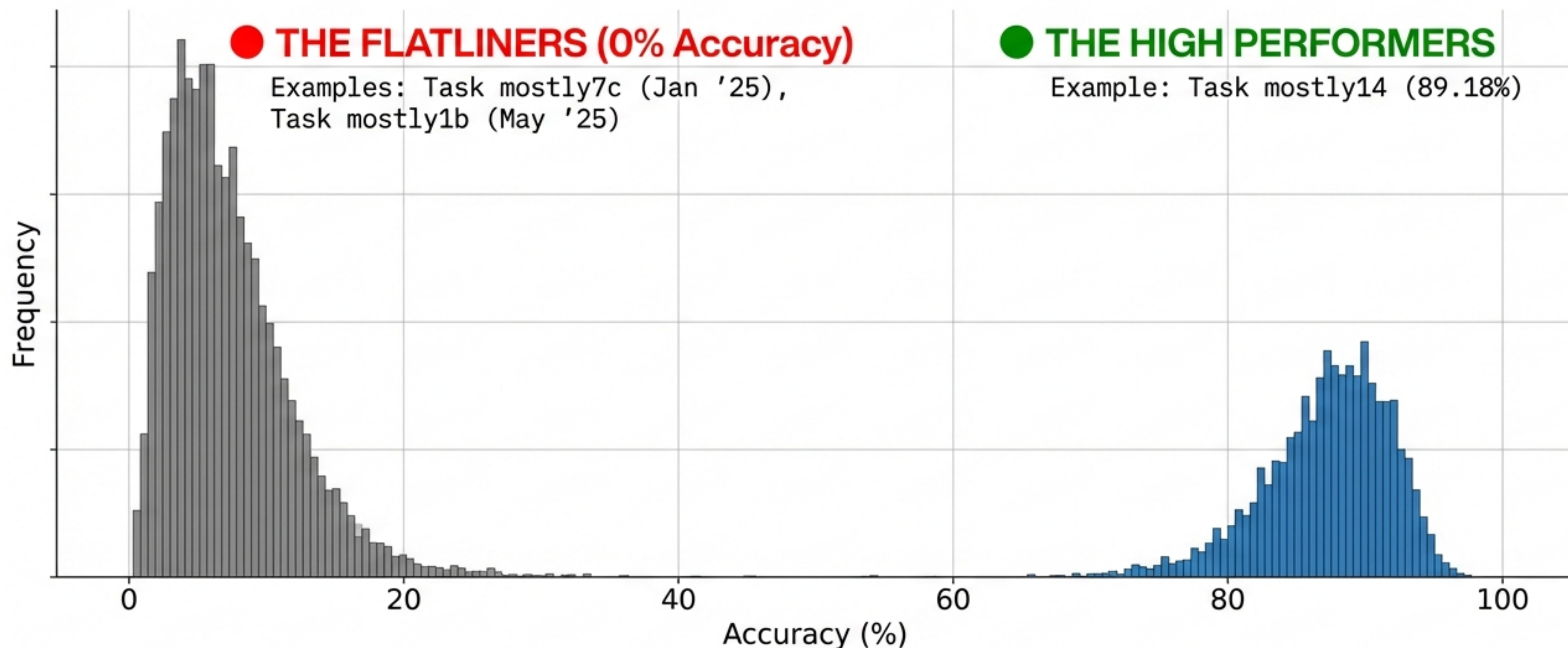


SYSTEM REFLEXES: THROUGHPUT VS. LATENCY



While the system is capable of sub-millisecond response, heavy loads push latency into the 300ms+ range.

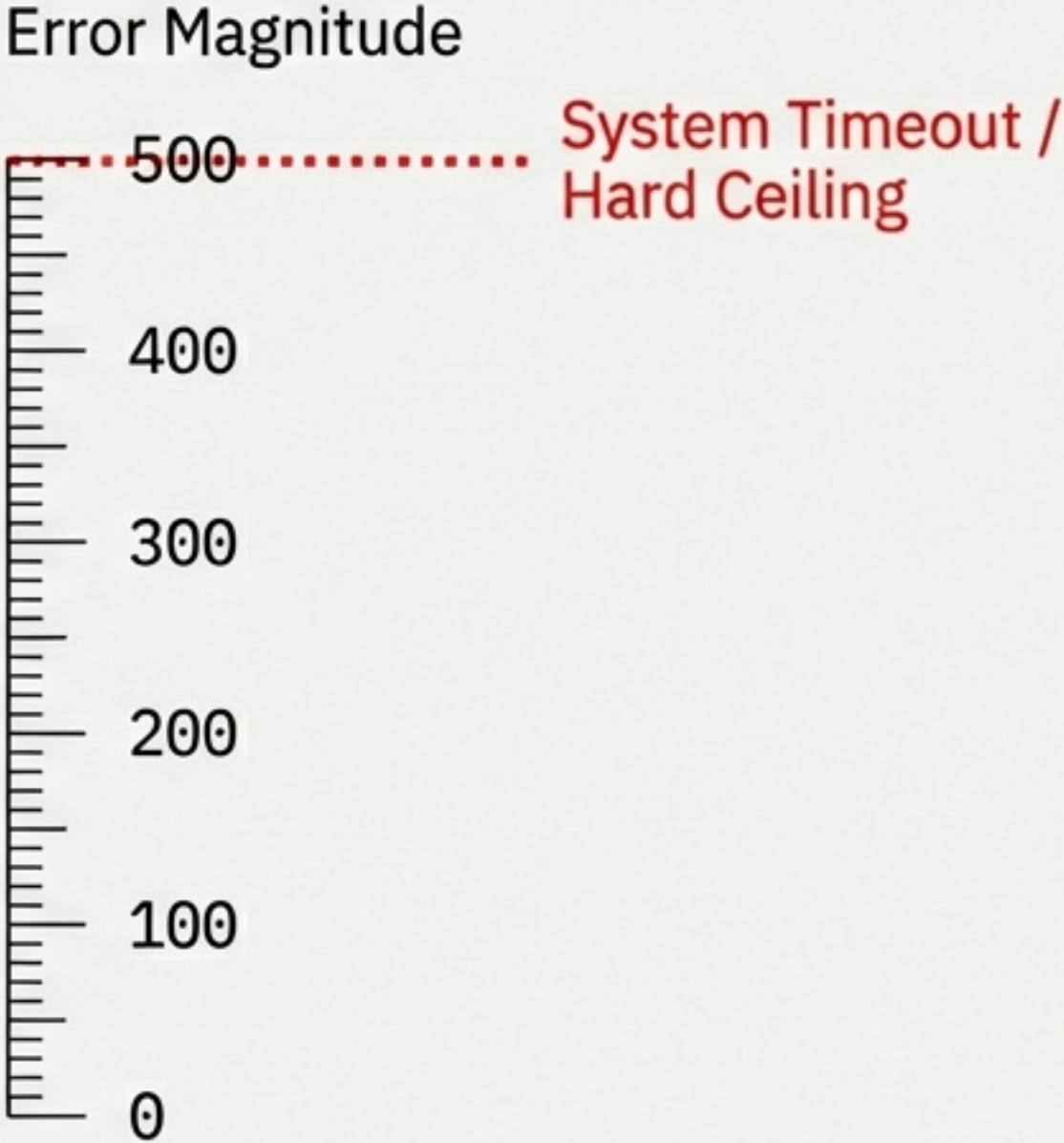
COGNITIVE FUNCTION: MODEL ACCURACY INSTABILITY



This split personality indicates that while some models are robust, a significant portion of compute is wasted on models that fail completely (0% accuracy).

CRITICAL FAILURES: THE '500' ERROR CEILING

TOP OFFENDERS (ERROR RATES)		
TASK ID	DATE	ERROR RATE
TASK ID: mostly21-69ef...	Date: June 2025	Error Rate: 499.372
TASK ID: mostly1e-9ab3...	Date: Oct 2024	Error Rate: 499.183
TASK ID: mostlya3-6779...	Date: June 2025	Error Rate: 170.853



We observe error rates capping out near 500, suggesting a hard timeout or max-retry failure rather than a calculated loss metric. Immediate audit of training pipelines required.

CASE STUDY: THE HEAVY LIFTER

SUBJECT: Task ID mostly29-d463-4688-904c-d571003ff7fd

TIMESTAMP: 14 Feb 2025, 06:46:03

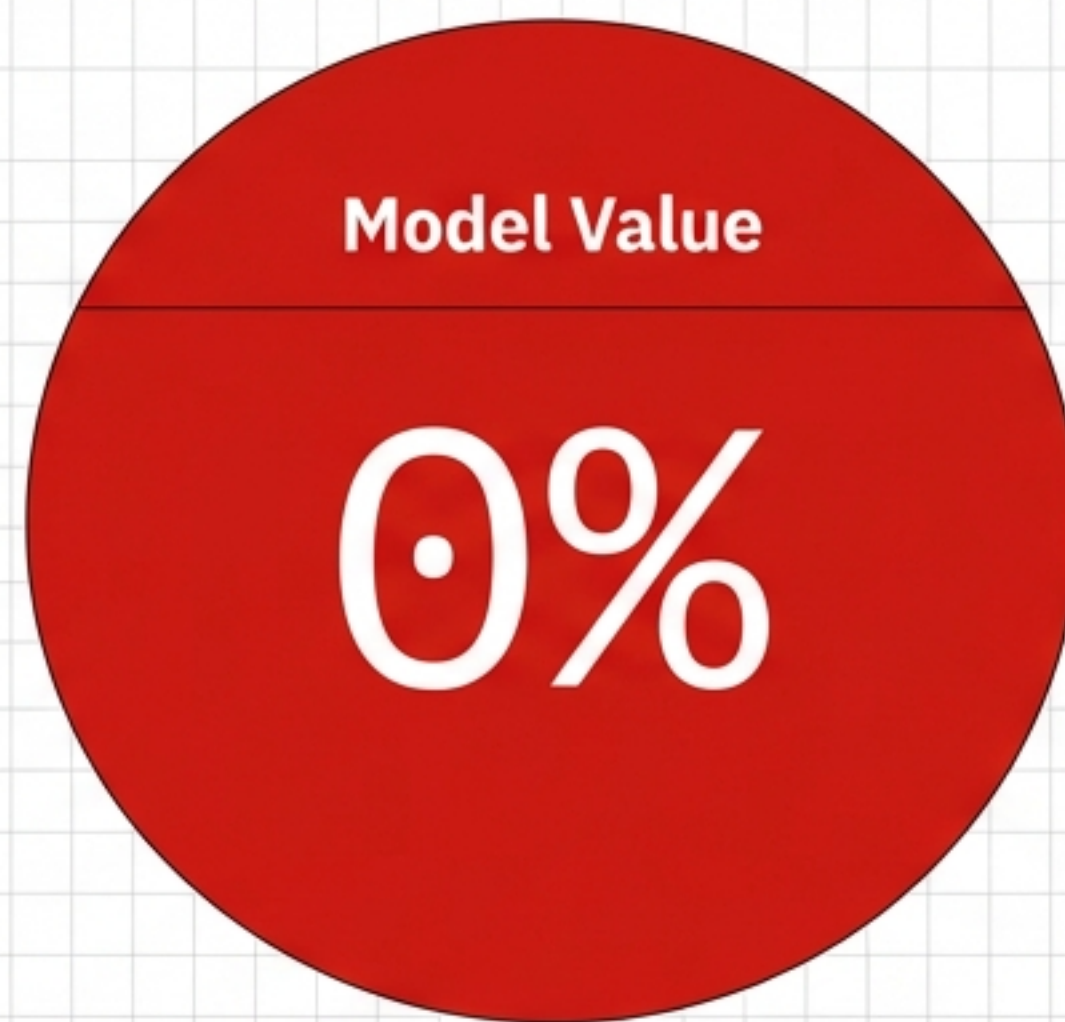
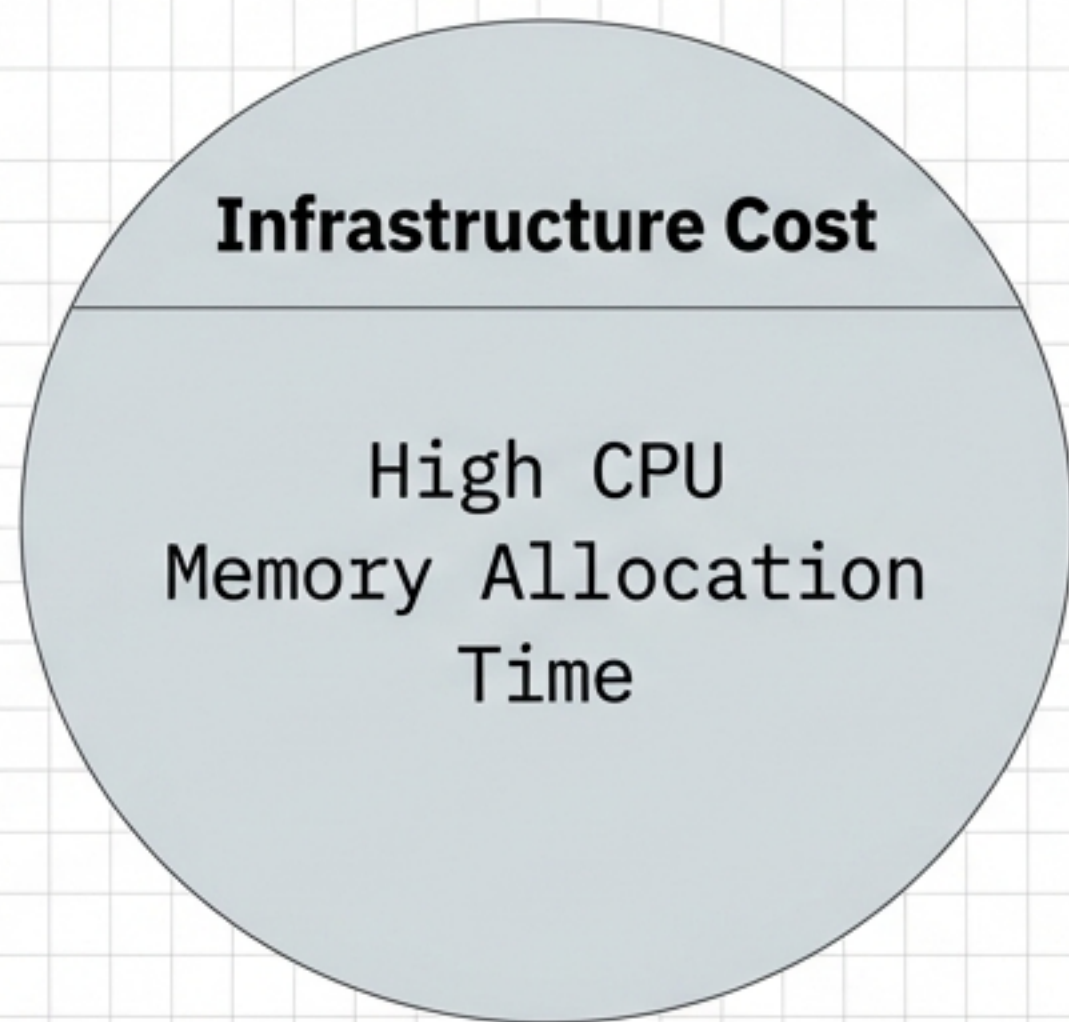
CPU USAGE: 1025.975%

CONTEXT: Cluster High-Activity Event

ASSESSMENT:

This task represents the highest recorded resource consumption. Likely a massive Graph Data Science (GDS) PageRank or Centrality calculation running across 10+ threads. While powerful, this task blocks other operations.

CASE STUDY: WASTED COMPUTE (THE SILENT FAILURES)



ZOMBIE MODEL EXAMPLES:

Example A: Task mostly1b-074a... (03 May 2025) -> Result: 0.007% Accuracy

Example B: Task mostly7c-6288... (21 Jan 2025) -> Result: 0.0% Accuracy

FINANCIAL IMPLICATION: These tasks represent 'Zombie Models'—consuming electricity and cloud credits to predict nothing.

ANOMALY DETECTION: THE OCTOBER 3RD EVENT

Event Timestamp: 2025-10-03 T 19:50:00

19:50:00	
Task mostly27...	CPU: 117%
Task mostly34...	CPU: 118%
Task mostlyfd...	MEMORY: 522.59
Task mostly2f...	GPU: 1032.4 (Error)
Task mostlyaa...	THROUGHPUT: 72.58
Task mostly07...	ACCURACY: 72.72%

SYNCHRONIZATION SPIKE.
Over 20 distinct tasks reported metrics at this exact second. This indicates a scheduled full-system stress test or batch ETL pipeline that saturated all layers of the stack simultaneously.

STRATEGIC RECOMMENDATIONS (THE PRESCRIPTION)

01 QUARANTINE 'ZOMBIE' MODELS

Implement early-stopping. Terminate any model with <10% accuracy after epoch 1.
(Ref: mostly7c)

02 FLATTEN THE CURVE

Implement a job queue to limit concurrent GDS projections. Prevent 1000%+ CPU spikes that destabilise the cluster.

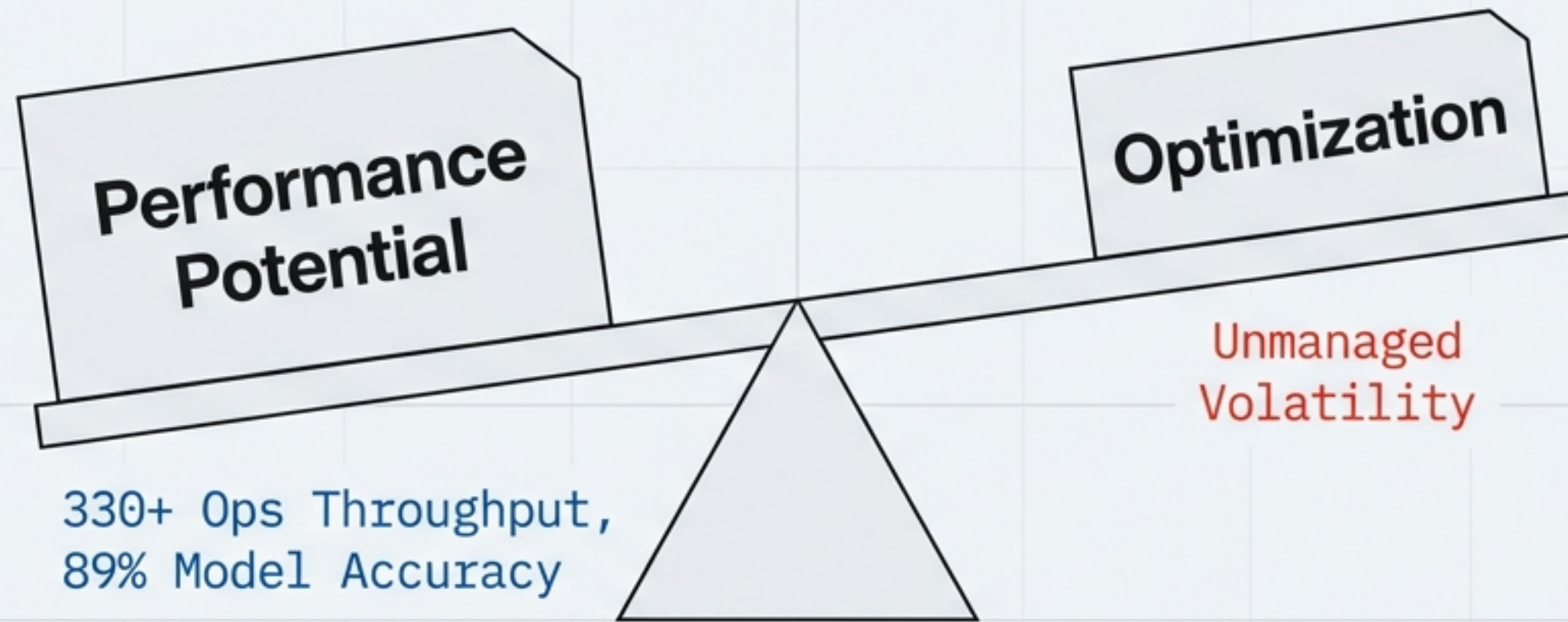
03 DEBUG GPU TELEMETRY

The '1032.4' value is a hard-coded error or overflow. Patch the metric collector to restore visibility.

04 INVESTIGATE OCT 3RD

Audit batch jobs from 03 Oct 2025 (19:50:00) to ensure this concurrency level does not occur during business hours.

PROGNOSIS: POWERFUL BUT UNREFINED



Optimising the bottom quartile of failed tasks could recover significant infrastructure capacity immediately.